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Lesson Notes

MODULE 8 | LESSON 1

COHERENT ASSET ALLOCATION AND CAUSAL STRUCTURE

Reading Time	20 minutes
Prior Knowledge	Bayesian network
Keywords	Econophysics

In the last module, we looked at different ways to incorporate a type of risk that superficially does not seem financial but in fact presents unprecedented risks to the financial system. We did this with and without Bayesian networks. In this module, starting with this lesson, we return to a near exclusive focus on Bayesian networks as a tool for risk financial management.

1. Portfolio Management Revisited: From the Other Side of the Coin

We have made the point before in this course that the Portfolio Management course already incorporated risk management, so naturally this Risk Management course will invoke many of the topics and themes from that course with a different emphasis.

For example, when D'Acunto et al. refer to the "significant level of redundancy between these factors, which might enhance risk contagion among multi-factor portfolios during financial crises" (1), they are referring to dangerous scenarios like the one we read about in Portfolio Management, of which the Quant Meltdown of August 2007 is a prime example (though it is not the only one). Recall from that previous course that investors believed that factor strategies were low risk, due to their liquidity and (potential) diversification, and so they leveraged the strategy. The leverage effectively multiplied the low risk levels, transforming them into much higher risk levels and opening the door for the resulting turmoil. In fact, the subprime mortgage crisis was still in its very early stages in the second half of 2007, so it did not require much of a "financial crisis" backdrop for the contagion to get out of control during that period. Remember it may have been losses on a mortgage subportfolio that caused a single asset manager (e.g., hedge fund) to sell from its factor-invested equity subportfolio. Think of this example when you read D'Acunto et al. discuss the link

between “the causal structure of equity risk factors” and “a worsening macroeconomic environment” (1).

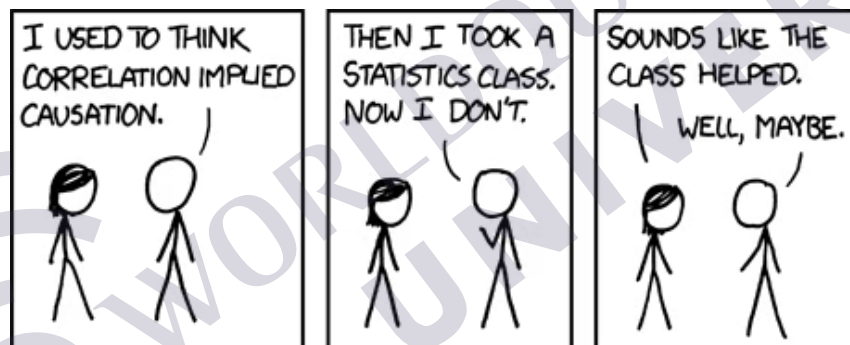
By the same token, remember de Prado’s appeal for more rigorous backtesting as discussed in “Backtesting without Overfitting: Deflating Sharpe and Overcoming Selection Bias” (Module 8, Lesson 1) when D’Acunto et al. describe the mechanism by which “model selection biases” help propagate an apparent zoo of “new” factors which are redundant, i.e., reformed or deformed versions of factors already fully exploited by the market (1).

1.1 Parenting as a Common Cause

On that note, D’Acunto et al. mention it, and we define here for you “Reichenbach’s Common Cause Principle: It is the claim that if two events are correlated, then either there is a causal connection between the correlated events that is responsible for the correlation or there is a third event, a so called common cause, which brings about the correlation” (*Rédei* 1).

This is not a very strong statement: It does of course capture that correlation between two variables does not necessarily imply that one causes the other, but it goes further by suggesting that either one causes the other *or* some other variable is causing the correlation. That other third variable of common cause would likely be modeled as a parent node to these two children nodes in a Bayesian network (though such node may be a hidden variable until it is observed, as has been discussed).

Figure 1: The Correlation between Correlation and Causality



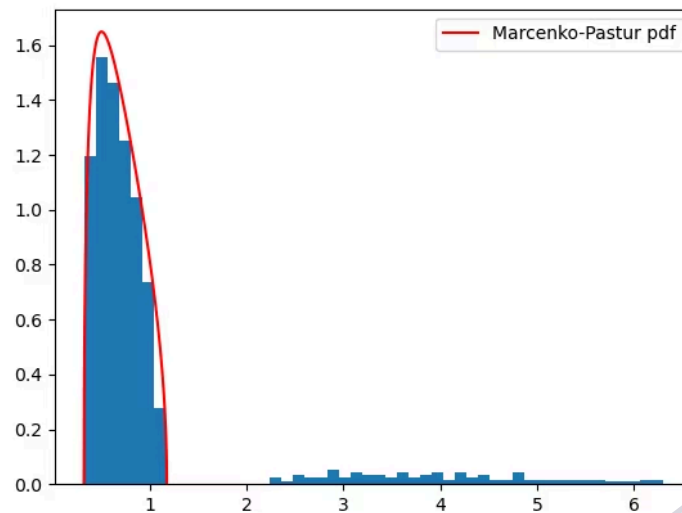
Source: XKCD. "Correlation." XKCD, <https://xkcd.com/552/>.

1.2 Eigenvalues as Common Cause

By the way, that common cause is often the most obvious one: D’Acunto et al. “provide evidence for the market factor being by far the dominant one” (8) especially in times of stress. Rebonato and Denev make related observations with respect to the role of correlation matrix eigenvalues: The market factor (or eigenvalue in this case) drives what is called the “instability” of returns (10). They criticize other authors for characterizing the larger eigenvalues (which include the market factor, of course) as “picking up noise” (Rebonato and Denev, pp. 10–13). Surely, you concur with Rebonato and Denev, given your work with de Prado’s “de-noising” technique, which we covered in Module 7, Lesson 2 of Portfolio Management. If so, you would prefer to say that the largest eigenvalues constitute a “tone” that a portfolio manager might choose to remove from the optimization exercise with a “de-toning” technique. These are intimately related but very different techniques and portrayals of the market factor that one needs to be clear about. In Portfolio Management, as here, we generally associate the smaller eigenvalues with noise as they tend to fit the Marcenko-Pastur distribution (MPD) so well, which, as you know, describes the distribution of eigenvalues for a

random matrix (see figure 2). Since such randomness does not provide information we can use to construct a more robust portfolio, we consider it noise.

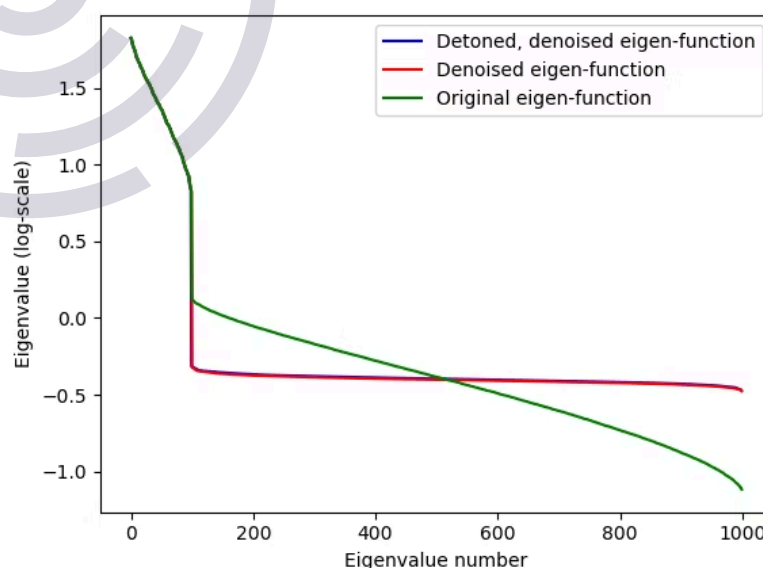
Figure 2: Histogram of Noisy and Signal Eigenvalues Fitted to Marcenko-Pastur Distribution



Source: Moen, Endre. "Machine_Learning_for_Asset_Managers." *GitHub*, https://github.com/emoen/Machine-Learning-for-Asset-Managers/tree/master/Machine_Learning_for_Asset_Managers.

In the figure, it is those eigenvalues that fall outside of the MPD that we consider information-rich signals. Nevertheless, there may be circumstances for which we prefer a detoned (in addition to denoised) correlation matrix, for which you find a comparative eigenvalue distribution plot in figure 3. De Prado elucidates the benefit of de-toning as follows: "By removing the market component, we allow a greater portion of the correlation to be explained by components that affect specific subsets of the securities. It is similar to removing a loud tone that prevents us from hearing other sounds" (31).

Figure 3: De-noising and De-toning Eigen-functions



Source: Moen, Endre. "Machine_Learning_for_Asset_Managers." *GitHub*, https://github.com/emoen/Machine-Learning-for-Asset-Managers/tree/master/Machine_Learning_for_Asset_Managers.

2. The Problems with Traditional Modern Portfolio Theory

Rebonato and Denev make the unsurprising claim (though they make it dramatically) that the “regularities” that so many models depend on break down “occasionally but violently” (2). It should come as no surprise that in the other required reading for this lesson, D’Acunto et al. reference “anomalies in times of financial turmoil” (2). Rebonato and Denev also make the point that portfolio construction models that are based on mean-variance optimization or a similar approach inherently equate volatility with risk, almost as if they were mere synonyms—and similarly “codependence becomes correlation” (2). Further, they denote that such models make little if any use of human expertise or domain knowledge, information that comes from experience beyond the data. Their solution to these issues with traditional approaches is to “provide an intuitively appealing, yet logically justifiable framework” (4) in the form of “a coherent portfolio optimization framework that is conditional on the current state of the world” (5). If any of these observations (for example, the word “conditional” in the previous sentence) made you think that we were turning our attention once again to the explanatory and predictive power of Bayesian networks, then you have certainly been paying attention.

Just as risk has been traditionally conflated with variance or volatility, it is often commingled with uncertainty itself in ways that are not helpful; thus, Rebonato and Denev remind us of Frank Wright’s distinction between the two: “according to Knight, risk applies to situations where we do not know the outcome of a given situation, but can accurately measure the odds. Uncertainty, on the other hand, applies to situations where we cannot know all the information we need in order to set accurate odds in the first place” (MIT). Rebonato and Denev stake a claim to the middle ground between these two concepts, seeking to exploit any and all information that might be useful for the portfolio optimization challenge at hand (4).

3. Conclusion

In this module, we explore Bayesian networks as a risk management tool alongside the use of factor analysis or factor investing. We saw that the latter also has important theoretical and empirical contributions to our understanding of the financial system as a network.

References

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